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DATA MINING AND BUSINESS INTELLIGENCE

Continuous Assessment 1

Case Study Based Analysis

Market Basket Analysis

Association Rules for Retail Sales and Merchandising Decisions

From Data to Decisions

Problem → Data → Preprocessing → Algorithm
→ Results → Evaluation → Insights → Decision

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Case study domain: Retail and sales analytics

Data Mining task: Association Rule Mining

Executive Summary and Report Structure

This case study examines how a grocery retailer can use transaction data to identify products purchased together and select practical merchandising experiments. The business problem is not simply to find the most popular products, but to discover combinations whose joint occurrence is stronger than their individual popularity would suggest.

The analysis uses the public Groceries case study, comprising 9,835 baskets and 169 product categories over 30 days [1]. Apriori and FP-Growth are justified as two suitable frequent-itemset algorithms. Association rules derived from their output are assessed using support, confidence and lift, together with stability and business feasibility checks.

The principal recommendation is to use credible rules to shortlist coordinated product placement, bundles or targeted cross-selling offers. These actions should be piloted against a comparison group. A positive association can support an experiment, but it does not establish that a discount or layout change will increase profit.

Available numerical findings are reported from the accompanying CA2 presentation [4] and labelled as reference results. The report separates those findings from proposed implementation checks and future business evaluation.

Report guide

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1 Problem and Business Objectives

1.1 Retail context and problem statement

A grocery retailer sells many categories and records numerous shopping baskets. Conventional sales summaries can show that milk or vegetables sell frequently, but they cannot by themselves reveal which products have useful co-purchase relationships. A product may appear in many combinations simply because it is popular. Treating every common combination as a special buying pattern can lead to ineffective offers.

Without systematic basket analysis, a manager may select bundles, adjacent displays and checkout recommendations mainly from experience. This can miss useful relationships or allocate discounts to items that customers would have purchased anyway. The selected problem is therefore to extract interpretable product associations from transaction history and use them to choose decisions worth testing.

1.2 Business questions

The analysis asks which itemsets occur frequently, which rules remain interesting after accounting for baseline popularity, how many baskets each rule reaches, and which relationships can support a practical merchandising action. It also asks what additional evidence is required before the retailer can claim a commercial benefit.

1.3 Objectives and intended outcomes

Analytical objectives: prepare reliable basket data; identify frequent itemsets; generate understandable association rules; compare suitable algorithms; and evaluate rule quality. Outputs include a cleaned transaction representation, itemset and rule tables, selected visualisations, and a shortlist of candidate actions.

Business objectives: improve cross-selling opportunities, help customers discover complementary products and increase basket value where commercially justified. Profit must be considered alongside revenue, because discounts, operating costs and substitution can eliminate apparent gains.

1.4 Stakeholders and decision boundaries

Store managers can review layout changes, merchandising teams can design offers, and analysts can document the evidence. Inventory staff should verify availability before a trial. Customers may benefit from more relevant offers or convenient displays, but the study does not establish individual customer preferences.

The decision supported by this case study is a limited merchandising pilot, not an automatic store-wide rollout. The retailer is treated as a realistic application context for a public dataset; no named supermarket partnership, actual deployment or measured increase in sales is claimed.

1.5 Definition of success

Analytical success means producing reproducible, non-redundant rules with adequate support and useful lift. Business success means that a controlled trial improves its predefined primary metric without unacceptable discount costs, stockouts or reductions in profit elsewhere.

2 Data Understanding and Source

2.1 Dataset provenance

The Groceries dataset distributed with the arules package contains 30 days of real point-of-sale baskets from a typical local grocery outlet. Its official documentation reports 9,835 transactions and 169 aggregated item categories [1]. These characteristics match the dataset described in the accompanying presentation [4]. The dataset is suitable because its unit of observation is already a shopping basket.

The official documentation is the canonical source for the case study. If a CSV mirror is used for CA2, its basket count, item categories and transaction boundaries should be checked against this description. The exact downloaded data file has not been inspected for this report, so no file-specific missing-value or duplicate counts are asserted.

2.2 Logical data dictionary

Element	Meaning	Role in the analysis
Basket identifier	A row index or transaction identifier	Separates one shopping event from another; not itself a mined product.
Item category	A product category such as yogurt or whole milk	The categorical entity that may appear in itemsets and rules.
Basket item list	All observed categories in one transaction	The input transaction set before encoding.
Item presence	True or false for a basket and category	The encoded feature used for frequent-itemset mining.

2.3 Granularity and interpretation

The analytical unit is the basket, not the customer. Several baskets could belong to the same person, but the transaction representation used here does not establish that relationship. Likewise, a product category is broader than a specific brand, pack size or stock-keeping unit. A category-level rule should not be translated into an exact brand recommendation without further evidence.

2.4 Available and unavailable information

The baskets support counts, product-presence frequencies and co-occurrence analysis. The supplied analysis does not establish item quantities, selling prices, product costs, customer identities or a usable timestamp for each basket. Consequently, revenue, margin, repeat-customer behaviour and time-of-day effects cannot be derived from item presence alone.

2.5 Fitness for the business problem

The data is appropriate for discovering candidate co-purchase patterns. Its short duration and single-outlet scope constrain generalisation. It can inform a hypothesis about product affinity, while retailer-specific operational data is needed to decide whether the resulting promotion is profitable and suitable for another store.

3 Exploratory Analysis and Data Quality

3.1 Initial inspection

Before mining, the analyst should verify that each record represents exactly one transaction. Inspect several raw rows, identify delimiters and quoting conventions, and check whether a header or basket identifier is present. An incorrect import can split one basket into multiple rows or treat metadata as a product, changing every downstream metric.

Count the number of non-empty baskets and distinct category labels. For the canonical dataset, compare these counts with 9,835 and 169 [1]. If they differ, investigate the source file and transformations before assuming that the difference is a harmless formatting issue. Record all exclusions so that the denominator used in support calculations remains clear.

3.2 Basket composition and frequency analysis

Examine the minimum, median and maximum number of categories per basket and plot a basket-size distribution. Very small baskets may offer limited opportunities for co-occurrence rules, while unusually large baskets should be inspected for parsing errors. A large basket should not be removed merely because it is uncommon; it may be a valid purchase.

Calculate each category's presence count and support. The presentation reports whole milk in 2,513 baskets, or approximately 25.6% of the 9,835 baskets [4]. This is a reference descriptive result. It illustrates why a frequent consequent can produce high confidence even when the relationship is not particularly distinctive.

3.3 Quality checks and handling principles

Check	Why it matters	Appropriate response
Blank product tokens	Create meaningless item labels	Remove empty tokens and inspect the source.
Inconsistent case or spaces	Split one category into several labels	Apply documented label normalisation.
Repeated product within a basket	May represent quantity or repeated entry	Encode presence once for this analysis.
Identical basket contents	May be separate valid transactions	Retain unless duplicate records are independently established.
Unexpected categories	May indicate import or mapping problems	Review against the category vocabulary.

3.4 Useful visualisations

An item-frequency bar chart explains product reach; a basket-size histogram explains transaction composition. A selected co-occurrence matrix can reveal candidates for deeper analysis, but raw pair counts must be evaluated against baseline popularity. Exploratory charts guide inspection; association metrics determine whether a pattern deserves further review.

4 Data Preprocessing

4.1 Cleaning transaction records

Load each transaction as a list of category labels. Remove blank tokens and trim surrounding whitespace. Standardise case if inconsistent naming is observed. Merge aliases only when they are confirmed to mean the same category; similar wording alone is insufficient. For example, two genuinely different dairy categories must not be combined merely to simplify the vocabulary.

Exclude empty baskets only after checking whether they result from missing data or a parsing problem. Keep a log of raw rows, excluded rows and final baskets. Treat a missing product value as unknown or absent according to its documented meaning; do not invent a product to complete a basket.

4.2 Repeated items and repeated baskets

This study models whether an item occurs, not how many units were purchased. Within a valid transaction, repeated occurrences of the same category can be reduced to one presence indicator. In contrast, two independent shopping events with identical products must remain as two baskets. Removing them because their contents match would bias product frequency and support.

4.3 Boolean transaction encoding

Convert the cleaned item lists into a Boolean transaction matrix. TransactionEncoder provides this representation in the planned Python workflow [5]. Each category becomes a column and each basket becomes a row. A true value means the category appears in that basket; false means it does not. The expected canonical shape is 9,835 by 169 when all baskets are retained.

Illustrative basket	Milk	Bread	Butter	Eggs
A	1	1	0	0
B	1	0	0	1
C	0	1	1	0
D	1	1	1	0

The table is an illustration of encoding, not a claimed extract from the original file. Item codes or category names should remain recoverable so that output rules can be explained in ordinary business language.

4.4 Validation after transformation

Check that every matrix value is Boolean, every column has a unique category label and the number of rows matches the documented retained transaction count. Recompute a few product counts directly from the basket lists and compare them with column totals. Check that no identifier or header has become an item.

4.5 Why conventional scaling is unnecessary

Numeric standardisation and min-max scaling are unnecessary because the features express membership. Label encoding a category as an arbitrary integer would also be inappropriate: the mining problem concerns sets of items, not numerical distances between category codes. A sparse representation may save memory when most product columns are absent in each basket.

5 Mining Task and Theoretical Foundations

5.1 Selection of Association Rule Mining

Association Rule Mining is an unsupervised descriptive task. It is appropriate because there is no predefined outcome label such as fraud or cancellation. The objective is to describe relationships among observed items. Classification would require a target to predict; clustering would group similar baskets or customers but would not directly produce the product implication rules required here.

5.2 Transactions itemsets and frequent patterns

Let I be the set of product categories and let each transaction be a subset of I. An itemset is any combination of categories. An itemset is frequent when the fraction of baskets containing all its items reaches a chosen minimum support. A frequent pair indicates sufficient joint occurrence, but its business usefulness still depends on the items' individual popularity and the decision context.

5.3 Meaning of an association rule

A rule A → B contains an antecedent A and a consequent B. Both are non-empty and they share no items. The rule describes how often B is present among baskets containing every item in A. For example, {curd, whole milk} → {yogurt} conditions on both antecedent categories being present, not either one individually.

The arrow is a conditional description rather than a causal or chronological statement. A reverse rule can have different confidence. Neither rule establishes which product entered the basket first, which item motivated the trip, or whether changing the display would create an additional purchase.

5.4 Core measures

Support: the proportion of all transactions containing the full combination. **Confidence:** the proportion of antecedent baskets also containing the consequent. **Lift:** confidence divided by the consequent's overall support. These measures capture reach, conditional reliability and association relative to baseline, respectively [6].

5.5 Two stages of the mining solution

The first stage discovers frequent itemsets using Apriori or FP-Growth. The second stage divides suitable itemsets into antecedents and consequents, calculates the rule metrics and retains rules satisfying the configured conditions. The algorithms discussed in the next sections solve the first stage; generating and selecting business rules remains a separate analytical step.

5.6 Practical interpretation

A high-confidence rule may simply recommend a product that is already common. A high-lift rule may apply to very few baskets. The analyst must consider both reach and affinity, then assess whether the relationship is stable, understandable and commercially actionable. No single measure replaces business judgement.

6 Apriori Algorithm and Justification

6.1 Main principle

Apriori discovers frequent itemsets through progressively larger candidates [2]. Its key pruning property is that every subset of a frequent itemset is also frequent. If a combination already fails the minimum-support requirement, adding more items cannot increase its support. Any larger candidate containing that infrequent combination can therefore be rejected.

6.2 Operational steps

Step 1: count all individual categories and retain those meeting minimum support. **Step 2:** join frequent single items to create candidate pairs. Count their occurrence and retain frequent pairs. **Step 3:** join compatible frequent pairs to create candidate triples, discarding any candidate with an infrequent subset. Continue until no larger frequent itemsets remain.

After itemset discovery, generate rules from itemsets containing at least two categories. Compute confidence for each eligible split and apply the confidence threshold. Calculate lift and supporting counts for the remaining rules. The mlxtend Apriori interface accepts the Boolean matrix used in this project [7].

6.3 Compact conceptual pseudocode

```
L1 ← frequent individual items
k ← 2
While L(k−1) is not empty
   Ck ← join compatible itemsets from L(k−1)
   Prune candidates with an infrequent subset
   Count the remaining candidates in all baskets
   Lk ← candidates meeting minimum support
   k ← k + 1
Return the union of all frequent itemset levels
```

6.4 Suitability for the selected case

Apriori is a good explanatory baseline for the Groceries problem because managers and examiners can follow how counts lead to itemsets and then rules. The eight-basket worked example in Section 10 makes its pruning logic transparent. Its output is a set of interpretable combinations rather than an opaque score.

6.5 Strengths and limitations

Strengths include a clear mathematical basis, explicit support thresholds and straightforward verification using transaction counts. Its main computational limitation is the number of candidates and repeated counting passes. When many categories are frequent or baskets overlap heavily, candidate generation can become expensive. Lowering support can substantially increase the search space.

6.6 Expected role in CA2

Implement Apriori as the baseline, export itemsets and rules, and record the thresholds. The report does not assign a measured runtime because execution logs were not supplied. Its suitability is justified by the problem structure and interpretability; its actual computational efficiency must be measured in the implementation environment.

7 FP Growth Algorithm and Justification

7.1 Main principle

FP-Growth discovers frequent patterns using a compressed transaction structure called an FP-tree [3]. Instead of explicitly enumerating the same sequence of candidate levels as Apriori, it uses shared prefixes and conditional pattern growth. It addresses the frequent-itemset task while offering a different computational strategy.

7.2 Tree construction

An initial scan counts item frequencies and identifies categories meeting minimum support. Items in each retained transaction are placed in a consistent frequency order. During another scan, the ordered transaction is inserted into the tree. Common prefixes share nodes and their counts increase. A header structure links occurrences of the same category.

7.3 Conditional pattern mining

The mining stage considers an item and collects the prefix paths associated with it, together with their counts. These paths form a conditional pattern base. A conditional tree is constructed where useful, and the process recursively extends patterns. Itemsets meeting support are returned and can then be passed to association-rule generation.

The original research motivates this approach through compression, pattern growth and decomposition into smaller conditional problems [3]. In the proposed Python workflow, the FP-Growth function can operate on the same encoded matrix used for Apriori [8].

7.4 Suitability for the selected case

Grocery baskets often contain overlapping categories, so shared structure may reduce repeated work. FP-Growth is a useful alternative when Apriori produces many candidates. Using the same input representation and support threshold also makes it practical to verify that both algorithms identify equivalent frequent itemsets.

7.5 Strengths and limitations

The method avoids explicit candidate generation and can benefit from transaction compression. However, the tree and conditional structures consume memory, and compression depends on the distribution of basket contents. When transactions share little structure, the advantage may be smaller. It is therefore inappropriate to claim a universal speed or memory benefit without measurement.

7.6 Interpretation of the output

FP-Growth does not make a rule more commercially valid simply because it is mined efficiently. Frequent itemsets must still be converted into rules, evaluated against baseline popularity and checked for feasibility. A high-quality implementation should preserve category names and return the support values needed for rule calculations.

7.7 Expected role in CA2

FP-Growth can be implemented as a second algorithm for comparison or selected as the main mining engine. The faculty requires at least one implemented algorithm for CA2, whereas CA1 requires justification of at least two suitable algorithms. This report supplies that justification for both approaches.

8 Algorithm Comparison and Configuration

8.1 Comparison on common criteria

Criterion	Apriori	FP-Growth
Discovery approach	Generate count and prune candidates	Compress baskets and grow conditional patterns
Main intermediate structure	Candidate and frequent itemset levels	FP-tree and conditional structures
Primary concern	Candidate explosion and repeated scans	Memory use and data-dependent compression
Explanatory value	Easy level-by-level demonstration	Shows an alternative scalable strategy
Comparable output	Frequent itemsets and support	Frequent itemsets and support

With identical input baskets, minimum support and equivalent restrictions, both algorithms should return the same frequent itemsets and support values. Output ordering can differ. Compare normalised itemsets and their values, rather than assuming that the first row from one method corresponds to the first row from the other.

8.2 Proposed baseline settings

The presentation's reference configuration uses minimum support 0.01 and minimum confidence 0.25 [4]. For 9,835 baskets, 1% is 98.35 baskets, so a qualifying itemset requires **at least 99 observed baskets**. A confidence threshold of 25% retains rules where at least one quarter of antecedent baskets contain the consequent; this is an initial filter rather than a business guarantee.

For positive cross-selling opportunities, inspect rules with lift above 1 and adequate supporting counts. Avoid selecting only the maximum lift. Consider shorter antecedents for interpretability and compare complex rules with simpler alternatives before retaining them.

8.3 Sensitivity analysis

A proposed support grid of 0.005, 0.01 and 0.02 can illustrate the reach-versus-complexity trade-off. A confidence grid such as 0.25, 0.40 and 0.60 can show how conditional reliability affects the shortlist. These values define a planned comparison; no additional run counts are claimed. Lower support generally admits more combinations, while stricter confidence retains fewer candidate rules.

8.4 Fair performance comparison

Run both methods on the same cleaned matrix, machine and software environment. Separate preprocessing time, itemset-mining time and rule-generation time. Use repeated timings under consistent conditions when feasible, and state whether memory is directly measured. If maximum itemset length is restricted for one method, apply the same restriction to the other.

8.5 Selection decision

Use Apriori to make the method transparent and FP-Growth to investigate an alternative computational strategy. Choose the operational method using measured efficiency and maintainability after confirming equivalent results. Algorithm speed should not displace the quality of the final rules or their business interpretation.

9 End to End Data Mining Process

Required process

Problem → Data → Preprocessing → Algorithm → Results → Evaluation → Insights → Decision

9.1 Problem to algorithm

Problem: identify product combinations that can support better merchandising. Define the analytical question and intended store action before mining, so the project does not end with an unexplained list of rules.

Data: use transaction-level grocery baskets. Document the canonical source, category granularity, observation period and known limitations. Inspect whether the downloaded representation preserves one shopping event per basket.

Preprocessing: clean tokens, standardise verified labels, preserve independent transactions and encode Boolean presence. Save the category mapping and transformation log. Validate matrix dimensions and a sample of counts.

Algorithm: apply Apriori and/or FP-Growth with recorded settings. Generate association rules from frequent itemsets, then calculate confidence, lift and supporting counts.

9.2 Results to decision

Results: produce interpretable itemset and rule tables. Visualise category popularity, the support-confidence trade-off and selected rule strengths. Preserve complete antecedents and consequents in every table.

Evaluation: assess rule metrics, stability, redundancy and feasibility. Where possible, recompute shortlisted rules on held-out transactions. Compare algorithm outputs before comparing execution times.

Insights: identify which associations have enough reach and distinctive affinity to justify a particular action. Explain how baseline popularity changes the interpretation of apparently reliable rules.

Decision: choose a small, feasible merchandising pilot, define a control condition and measure the outcome. Expand or stop the intervention based on evidence. Feed operational findings back into later analysis.

9.3 Reproducibility and project outputs

The CA2 project should retain the original data, preprocessing notebook, package versions, thresholds, exported rule table and charts. Record each run's date, input transaction count and any itemset-length restriction. Supporting counts should be available for a manual verification of selected rules.

The minimum implementation deliverable is one correctly executed mining algorithm with evaluated results, suitable visualisations and business interpretation. The second algorithm is valuable for comparison, while this CA1 report provides the analysis and justification required before or alongside implementation.

10 Worked Transaction Example

The following eight-basket example from the presentation illustrates the calculations and Apriori logic [4]. It is a teaching example, separate from the full Groceries dataset.

ID	Items
1	Bread Butter Milk
2	Bread Butter
3	Bread Milk Eggs
4	Bread Butter Milk Jam
5	Milk Eggs
6	Bread Butter Jam
7	Bread Milk
8	Butter Milk Eggs

10.1 Itemsets with minimum support 37.5 percent

A support threshold of 3/8 requires an itemset to occur in at least three baskets. Bread occurs 6 times, Milk 6, Butter 5 and Eggs 3; all are retained. Jam occurs twice and is excluded. Among pairs, Bread + Milk occurs 4 times, Bread + Butter 4, Butter + Milk 3 and Eggs + Milk 3. Bread + Eggs and Butter + Eggs occur once each and fail the threshold.

Bread + Butter + Milk is a valid candidate because its three pairs are frequent, but its count is only 2, so it is rejected. Any larger itemset containing an infrequent subset is pruned. This example shows that a candidate must still be counted even when its subsets are frequent.

10.2 Rule calculations

Rule	Support	Confidence	Lift
Eggs → Milk	3/8 = 37.5%	3/3 = 100%	1 / 0.75 = 1.33
Butter → Bread	4/8 = 50%	4/5 = 80%	0.80 / 0.75 = 1.07
Bread → Milk	4/8 = 50%	4/6 = 66.7%	0.667 / 0.75 = 0.89

10.3 What the example teaches

Eggs → Milk has perfect confidence in only three antecedent baskets, so its apparent reliability rests on a small sample. Butter → Bread has high confidence but lift only slightly above 1. Bread → Milk exceeds 60% confidence, yet its lift is below 1 because Milk is already very common.

With a 60% minimum confidence, all three displayed rules pass that filter. A positive-affinity filter based on lift would reject Bread → Milk. These calculations demonstrate why confidence alone is insufficient and why small-sample rules should be validated before business use.

11 Available Results and Visual Interpretation

11.1 Status of the available evidence

The accompanying presentation reports a reference Groceries analysis using minimum support 1% and confidence 25%, producing 170 association rules [4]. It also reports whole milk in 2,513 baskets. These are available reference findings rather than a new execution performed for this report. Their reproduction should be checked against the actual CA2 dataset, parameters and exported rule table.

11.2 Selected reference association rules

Antecedent → consequent	Support	Confidence	Lift
Citrus fruit + other vegetables → root vegetables	1.04%	35.9%	3.30
Other vegetables + tropical fruit → root vegetables	1.23%	34.3%	3.14
Beef → root vegetables	1.74%	33.1%	3.04
Citrus fruit + root vegetables → other vegetables	1.04%	58.6%	3.03
Curd + whole milk → yogurt	1.01%	38.5%	2.76

Values are rounded as reported in the presentation. Rounded percentages should not be used to invent exact supporting counts; counts should be taken from the original transaction data.

11.3 Interpretation of the leading rule

For citrus fruit + other vegetables → root vegetables, confidence 35.9% means that approximately 35.9% of baskets containing both antecedent categories also contain root vegetables. Lift 3.30 means this conditional proportion is about 3.3 times the overall root-vegetable proportion. It does not mean sales would rise by 230% after a promotion.

Support 1.04% indicates that the entire combination occurs in a small fraction of all baskets. A focused offer or display test may therefore be more appropriate than a blanket store-wide discount. The potential audience and margin must be checked before selecting an intervention.

11.4 How visualisations should be read

A frequency chart highlights popular categories, not necessarily distinctive associations. A support-confidence scatter plot shows reach and conditional reliability; lift can be annotated directly to preserve a black-and-white presentation. A lift-ranked chart compares relative affinity but should display support alongside the ranking.

A network graph can summarise related products, but a rule with two antecedent items must not be interpreted as two independent pairwise rules. Tables remain necessary to preserve the full conditional statement. Visualisations help explain the results; they do not replace the underlying counts or validation.

12 Evaluation Measures and Validation

12.1 Core metrics

Let N be all baskets, count(A) the baskets containing every item in A, and count(A ∪ B) those containing the combined itemset. The principal measures are defined below [6].

Measure	Definition	Question answered
Support	count(A ∪ B) / N	How common is the whole combination?
Confidence	support(A ∪ B) / support(A)	How often does B accompany A?
Lift	confidence(A → B) / support(B)	How strong is association relative to baseline?

Lift above 1 indicates positive association; 1 indicates independence; below 1 indicates negative association. Support is a reach measure, and confidence is directional. None of the three measures directly quantifies profit or the causal effect of an offer.

12.2 Additional quality checks

Leverage equals support(A ∪ B) − support(A) × support(B). It expresses the excess joint occurrence above the independence baseline in absolute probability terms. It can complement lift when comparing a rare, high-lift combination with a broader opportunity.

Coverage and redundancy: inspect antecedent counts and the number of baskets supporting each rule. Compare complex rules with simpler ones so that extra antecedents are retained only when useful. Reject rules that cannot be acted upon because products are unavailable or the required discount is commercially unattractive.

12.3 Validation and robustness

If a discovery/validation split is used, mine candidates on the discovery subset and recompute their metrics on validation baskets. Do not tune repeatedly on the validation subset and then describe it as untouched evidence. A random split tests stability across sampled baskets, but it does not demonstrate future-season performance. A temporal split requires actual transaction timestamps.

Examine sensitivity to support and confidence settings. Changes in the top rules should be explained rather than hidden. A large search can reveal chance patterns; independent validation and, where feasible, resampling or statistical checks can strengthen the evidence. Generating more rules is not itself a quality measure.

12.4 Efficiency and business evaluation

Measure mining runtime separately from cleaning and rule generation. Compare itemset equivalence before efficiency. Standard classification accuracy is not the primary metric because this project describes associations rather than predicting a labelled target.

The final business evaluation is a controlled trial using a predefined metric such as average basket value or profit per eligible transaction. Report the size and uncertainty of any improvement, as well as operating costs and stock availability. Basket relationships alone cannot establish incremental business impact.

13 Business Insights and Decision Framework

13.1 Translating the reference findings into actions

The reference rules suggest produce-related opportunities and a dairy cross-selling relationship [4]. These findings should generate testable proposals rather than automatic discounts. The rule identifies where an opportunity may exist; feasibility and controlled evaluation determine whether the retailer should act.

Evidence	Candidate action	Primary check
Produce combinations associated with root vegetables	Pilot coordinated produce placement or a targeted recommendation	Compare incremental basket value with a control.
Curd + whole milk associated with yogurt	Offer an eligible-basket yogurt recommendation or coupon	Compare profit after discounts and redemption costs.
High popularity can inflate confidence	Require adequate support and lift before shortlisting bundles	Reject weak or unstable rules despite high confidence.

13.2 Proposed pilot design

Select one feasible action and define eligible transactions before the pilot. Assign comparable stores, customers or transactions to intervention and control where operationally practical. For layout changes, store-level or carefully managed time-block designs may be more appropriate than transaction-level assignment. Avoid exposing the control group to the same intervention.

Keep other major promotions as comparable as possible, monitor stock availability, and define the trial duration and analysis method in advance. Use a duration justified by transaction volume and the effect the retailer needs to detect; the case study does not establish a universal number of trial days.

13.3 Measures that connect analysis to business value

Average basket value is revenue divided by transaction count. **Bundle attach rate** is the fraction of eligible transactions containing the promoted combination. **Items per basket** requires unit quantities if interpreted as units; the Boolean data can instead measure distinct categories per basket. **Profit per eligible transaction** requires prices, costs and discount information.

Check cannibalisation: additional yogurt sales may replace another profitable product rather than create new demand. Also check whether discounted bundles mainly benefit customers who would have purchased the combination at full price. These effects can make a popular offer unprofitable.

13.4 Decision rule and monitoring

Expand an intervention only if the primary outcome improves sufficiently and operational checks remain acceptable. Revise or stop it when the evidence is weak or costs offset the benefit. After rollout, monitor changes in basket composition and refresh rules periodically. Historical affinity can change when prices, availability or customer behaviour change.

14 Limitations Future Scope and Conclusion

14.1 Data and methodological limitations

The public data represents one outlet over one month, limiting conclusions about other stores, regions or seasons. Category aggregation hides brands and pack sizes. The supplied basket representation does not establish quantities, margins, timestamps or customer identity. These gaps constrain commercial valuation, sequencing and personalised targeting.

Rule discovery is threshold-dependent. Low support may produce many fragile combinations, while high support can omit worthwhile niche patterns. High lift can arise for rare outcomes, and high confidence can reflect a popular consequent. Evaluating many candidate rules also increases the chance of finding patterns that do not persist.

Association does not establish causation. Existing placement, promotions or stock constraints may explain co-purchases. A rule cannot tell the manager how much additional demand a discount will create. The full-dataset figures in Section 11 remain reference findings until reproduced in the CA2 notebook.

14.2 Responsible use

Aggregate basket patterns can support merchandising without identifying individuals. Customer-specific targeting requires an appropriate basis for using customer data and clear limits on access and retention. Avoid inferring sensitive personal characteristics from a grocery combination. Recommendations should be presented as suggestions, with their evidence and limitations understood by decision-makers.

14.3 Future extensions

Add price, margin and inventory information to rank actions by commercial value rather than affinity alone. Compare rules across stores and seasons when suitable data becomes available. Customer-segment analysis can investigate whether aggregate patterns conceal meaningful differences, provided the required data is available and appropriately used.

Sequential pattern mining could study purchase order across transactions when timestamps and customer linkage exist. High-utility itemset mining could incorporate quantities and profit. Incremental or streaming mining could help refresh patterns as new baskets arrive. These are extensions rather than claims about the current implementation.

14.4 Conclusion

Market Basket Analysis is a suitable solution to the selected retail problem because it turns transaction records into understandable product relationships. Apriori and FP-Growth provide two justified approaches to frequent-itemset discovery. Careful preprocessing preserves the meaning of baskets, while support, confidence, lift and validation help distinguish useful associations from simple popularity.

The final business recommendation is to shortlist a small number of feasible product-placement, bundle or cross-selling actions and test them with an appropriate comparison group. The complete chain is therefore satisfied: problem definition leads to data preparation, mining, evaluation, insight and a measurable decision. Commercial improvement is the objective to verify, rather than an outcome assumed from the rules.

References and Assessment Coverage

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Web references accessed 29 October 2026. Software interfaces may vary by installed version; the implementation should record its package versions.

Coverage of the CA1 brief

Faculty requirement	**Report location**
Problem and business objective	Section 1
Data understanding and source	Sections 2 and 3
Data preprocessing	Section 4
Appropriate Data Mining task	Section 5
At least two suitable algorithms and justification	Sections 6 7 and 8
Complete Data Mining process	Section 9
Expected or available results and evaluation measures	Sections 10 11 and 12
Insights and support for the final business decision	Sections 13 and 14